INCH POUND

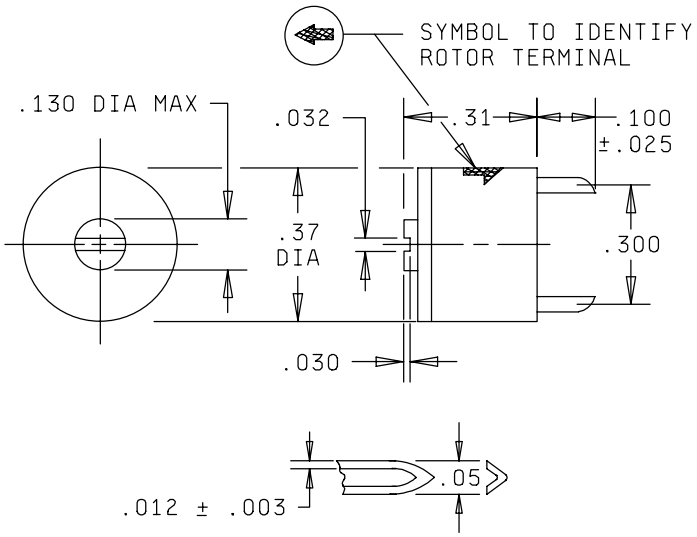
MIL-PRF-81/4E
21 April 2006
SUPERSEDING
MIL-C-81/4D
25 May 1999

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, VARIABLE, CERAMIC DIELECTRIC,
STYLES CV31 AND CV32

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-81.

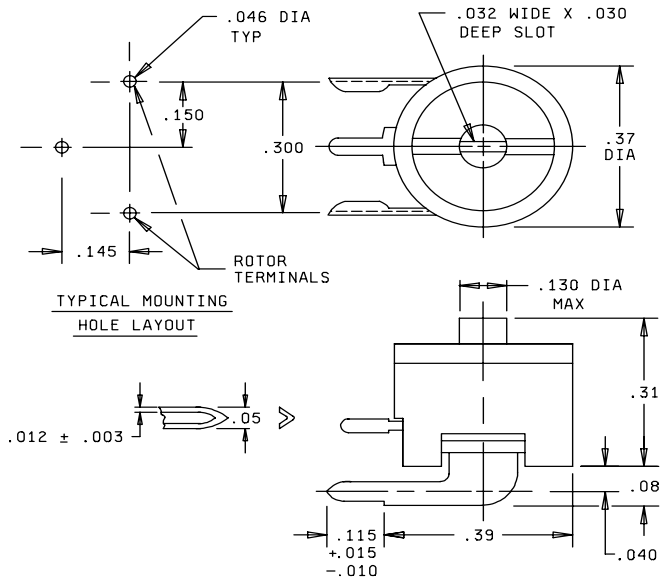



Style CV31

FIGURE 1. Dimensions and configurations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | .08 | .08 | 2.03 |
| .01 | .25 | .100 | 2.54 |
| .012 | .30 | .115 | 2.92 |
| .015 | .38 | .130 | 3.30 |
| .025 | .64 | .145 | 3.68 |
| .030 | .76 | .150 | 3.81 |
| .032 | .81 | .300 | 7.62 |
| .040 | 1.02 | .31 | 7.87 |
| .046 | 1.17 | .37 | 9.40 |
| .05 | 1.27 | .39 | 9.91 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.51 mm) for two-place decimals and ±.010 (0.25 mm) for three-place decimals.

Style CV32

FIGURE 1. Dimensions and configurations - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristics: See table I.

TABLE I. Styles CV31 and CV32.

| Type designation 1/ | Capacitance (pF) | | DC rated voltage (volts) | Characteristics | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Symbol | Capacitance change from value at +25°C | | | |
| | | | | | At -55°C | | At +125°C | |
| | Minimum | Maximum | | | Minimum percent | Maximum percent | Minimum percent | Maximum percent |
| CV3_A080 | 2.0 | 8.0 | 350 | A | -4.5 | +2.0 | -2.5 | +2.0 |
| CV3_A180 | 5.5 | 18.0 | 350 | A | -4.5 | +2.0 | -2.5 | +2.0 |
| CV3_B110 | 2.5 | 11.0 | 350 | B | -1.0 | +3.5 | -2.5 | -0.5 |
| CV3_B250 | 7.0 | 25.0 | 350 | B | -1.0 | +3.5 | -2.5 | -0.5 |
| CV3_C100 | 3.0 | 10.0 | 350 | C | -1.0 | +6.5 | -4.0 | -1.0 |
| CV3_C250 | 8.0 | 25.0 | 350 | C | -1.0 | +6.5 | -4.0 | -1.0 |
| CV3_D150 | 3.0 | 15.0 | 200 | D | +1.5 | +8.2 | -5.0 | -1.5 |
| CV3_D350 | 9.0 | 35.0 | 200 | D | +1.5 | +8.2 | -5.0 | -1.5 |
| CV3_E600 | 15.0 | 60.0 | 200 | E | +3.0 | +14.0 | -10.0 | -3.0 |

1/ The complete type designation will include an additional digit to indicate style CV31 or CV32.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. A dc potential of 2.2 times rated voltage applied between terminals for 3 seconds ±2 seconds.

Temperature coefficient: Within the limits specified for the applicable characteristic.

Capacitance drift: Within 0.75 percent of initial step 1 measurement or 0.50 picofarad (pF), whichever is greater.

Terminal strength:

Pull test: Capacitor held by body and 4-pound-load applied to each terminal for at least 10 seconds.

Torque: Not less than 1 ounce-inch nor more than 6 ounce-inches.

Barometric pressure (reduced):

Method 105 of MIL-STD-202, condition D (100,000 feet).

Test potential: 100 percent of dc rated voltage.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.

Insulation resistance:

Method 302 of MIL-STD-202, condition A, 100 volts dc applied: 10,000 megohms, minimum.

Capacitance:

Method 305 of MIL-STD-202.

DF: At 1 MHz ±100 kHz, at maximum and minimum capacitance: Shall be not more than 0.2 percent.

Fatigue:

ΔC: Shall not exceed 12 percent or 0.75 pF, whichever is greater.

Torque: Not less than 1 ounce-inch nor more than 10 ounce-inches.

Life:

Qualification test: 1,000 hours at +85°C, 150 percent of rated volts dc with a peak alternating voltage of 50 percent of rated volts dc (100 hertz or less) superimposed.

Insulation resistance: Initial requirement.

Capacitance change: Shall not exceed ±8 percent of initial value or 0.5 pF, whichever is greater.

Group C life: Conditions and requirements are the same as that required for qualification.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's).

Vibration, high frequency:

Method 204 of MIL-STD-202, condition B (15 g's).

Capacitance change: Shall not exceed ±2 percent or 0.5 pF, whichever is greater.

DF: Shall be not more than 0.2 percent.

A dc potential of 2.2 times rated voltage applied between terminals for 3 seconds ±2 seconds.

Insulation resistance:

10,000 megohms, minimum.

Moisture resistance:

Method 106 of MIL-STD-202:

Insulation resistance: 10,000 megohms, minimum.

Capacitance change: Shall not exceed ±5 percent of nominal value or 0.5 pF, whichever is greater.

DF: Shall be not more than 0.5 percent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.

CHANGES FROM PREVIOUS ISSUE: The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* Reference documents. In addition to MIL-PRF-81, this document also references MIL-STD-202.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activity:
Air Force - 19

Preparing activity:
DLA - CC

(Project 5910-2005-032)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.